AHMED NASSER BASSUIONY

Artificial Intelligence & Data Engineer

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PROFESSIONAL SUMMARY

Results-oriented AI, Machine Learning & Data Engineer with a Dual Bachelor's in AI and international academic experience at Universidad Autónoma. Expert in designing data pipelines and building end-to-end solutions, including PySpark notebooks for large-scale data processing. Proficient in YOLO-based computer vision and FastAPI. Experienced in managing structured and unstructured data to support advanced AI research and reliable production workflows.

TECHNICAL SKILLS

- **Data Engineering**: PySpark (Notebooks), Data Pipeline Architectures (ETL/ELT), Microsoft Fabric (Familiarity), Data Lakes, Dataflows, Structured & Unstructured Data Processing.
- **Software Developmen**t: Object-Oriented Programming (OOP), Software Design Patterns, System Integration, Git/Version Control, Documentation & Technical Writing.
- **Cloud & Automation**: Microsoft Azure, AWS (Basic), Docker, GitHub Actions, Linux (PowerShell/Bash), System Monitoring & Troubleshooting.
- **Robotics & Systems**: ROS 2 (Humble/Foxy), SLAM (Mapping), Path Planning, Gazebo Simulation, Sensor Fusion (Lidar/IMU), Actuator Control.
- **AI & Machine Learning**: TensorFlow, PyTorch, YOLOv8 (Object Detection), Computer Vision (OpenCV), Natural Language Processing (NLP), Scikit-learn.
- **Programming Languages**: Python (Expert), SQL, C++, C, R.

WORK EXPERIENCE

Data Engineer (Intern & Junior) | COB Solutions (July 2024 – Dec 2024)

- **Data Infrastructure**: Architected and maintained scalable ETL pipelines to transform raw, unstructured data into reliable datasets for ML workflows.
- **Workflow Automation**: Leveraged Airflow to automate complex data workflows and GitHub Actions for CI/CD, ensuring consistent data delivery.
- **Cloud & Warehousing**: Optimized data storage and modeling in Snowflake and GCP, utilizing SQL for modern data warehousing solutions.
- **Reliability**: Monitored and documented system performance to ensure the integrity of the data infrastructure for cross-functional teams.

Robotics Developer (RoboCup@Home Education) | Eindhoven, Netherlands (June 2024)

- **System Integration**: Architected communication between sensors and actuators using ROS 2 nodes, topics, and services to ensure reliable data exchange.
- **Autonomous Navigation**: Implemented SLAM for environment mapping and real-time path planning for precise obstacle avoidance.
- **Perception & Vision**: Utilized computer vision libraries to perform object detection, facial recognition, and person tracking in dynamic settings.
- **Human-Robot Interaction (HRI)**: Integrated Natural Language Processing (Speech-to-Text/TTS) to enable verbal command execution.
- **Simulation & Testing**: Developed and validated robot behaviors in high-fidelity 3D Gazebo environments to troubleshoot systems before physical deployment.

Machine Learning Intern | Telecom Egypt (WE) **(**March 2023)

- **Predictive Modeling:** Collaborated on machine learning models to analyze telecommunications data and improve service efficiency.
- **Exploratory Data Analysis (EDA)**: Utilized Python (Pandas/NumPy) to perform statistical analysis on large-scale network datasets.

Programming Instructor | iSchool (July 2024 – DEC 2024)

- **Curriculum Delivery**: Taught programming fundamentals to students using visual and AI tools; delivered lessons via Code.org, Scratch, and Pictoblox.
- **Mentorship**: Mentored 60+ students in developing logic and creative programming skills through project-based learning.

Programming Instructor | Eyoth (Feb 2026 – Present)

- **Educational Training**: Facilitated coding workshops focusing on foundational logic and software development principles for early learners.
- **Documentation:** Created custom lesson plans and documentation to track student progress and learning outcomes.

TECHNICAL PROJECTS

IntelliLearn | Multimodal AI Data Processing Platform

- **Automated Data Ingestion**: Architected a FastAPI backend to ingest and preprocess diverse data formats, including audio, video, and PDF, utilizing FFmpeg for standardized format conversion.

- **Unstructured Data Pipelines**: Developed an end-to-end pipeline using the Whisper model for high-fidelity audio transcription and speaker diarization, transforming unstructured audio into searchable text.

- **Text Processing & Chunking**: Engineered a robust PDF processor that extracts, parses, and paginates content into token-limited chunks, ensuring compatibility with LLM context windows via the Groq API.

- **System Reliability**: Implemented a modular system architecture with automated testing via Pytest and secure environment management to ensure consistent performance across development and production.

Quranic NLP Analysis | Semantic & Text Processing

- **Large-Scale Data Cleaning**: Processed a dataset of 6,200+ records of structured and unstructured religious text, performing extensive tokenization and cleaning to ensure high data quality.

- **Semantic Modeling**: Applied NLP techniques to analyze semantic relationships within the text, demonstrating the ability to extract meaningful insights from complex data structures.

- **Reliability**: Built a repeatable processing workflow to maintain data integrity across multiple iterations of the analysis.

Personal Portfolio | Full-Stack Technical Showcase

- **Web Architecture**: Engineered a responsive personal portfolio using modern web technologies to host and showcase large-scale AI and Data Engineering projects.

- **Cloud Deployment**: Configured and managed the deployment of the site, ensuring high availability and optimized loading times for technical project documentation and media.

- **Technical Communication**: Translated complex system architectures and data flows into clear, visual documentation for global recruiters and technical leads.

EDUCATION

Bachelor of Science in Artificial Intelligence | **(2020 – 2024**)

Faculty of Artificial Intelligence | Arab Academy for Science and Technology (AAST)

- **Department:** Intelligent Systems Department.
- **International Dual Degree:** Universidad Autónoma de Madrid (UAM), Spain.
- **UAM Summer Semester (International Intensive):** Completed specialized coursework in Data Warehousing, Robotics & Control Systems, and Database Management Systems (SQL).

AWARDS & CERTIFICATIONS

- **RoboCup@Home Education Certificate | LaTTE (Learning and Teaching Technology Environment)**: Issued for completing advanced robotics modules and participating in the Eindhoven, Netherlands competition.

- **Artificial Intelligence Internship Certificate | Telecom Egypt (WE)**: Validated professional experience in machine learning and data analysis infrastructure.

- **RoboCup Junior Participation | Line Tracking Competition**: Recognized for engineering autonomous sensor-based navigation systems.

- **Dual Degree Excellence Award | Universidad Autónoma de Madrid**: Academic recognition for performance in the international partner program.

LANGUAGES & TOOLS

- **Languages:** Arabic (Native), English (Professional Working Proficiency), Spanish (Elementary).

- **Environment:** Linux (Ubuntu), Windows (PowerShell/WSL), Docker, Git/Version Control.